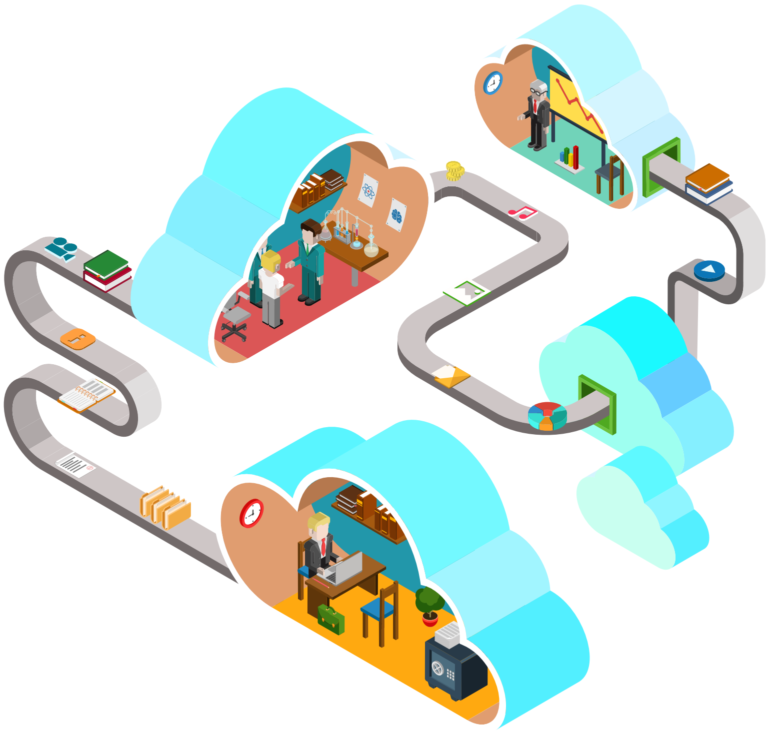
Managing Multiple Clouds with Open Source Tools



There are various challenges associated with management of services from multiple clouds. These include the variety of security services, operational tools, monitoring services and inconsistency between cloud platforms for core services like compute, storage and networking. However, there are a few open source cloud management tools that help to address these challenges.

Organisations prefer to use multi-cloud management solutions for a number of reasons.

1. During cloud migration, depending on the tech stack of applications and the migration approach (re-factor instead of re-architect), a multi-cloud solution can help as each cloud provider can enable different applications to be migrated faster.
2. During mergers and acquisitions, different organisations have application workloads with different cloud providers, and hence a multi-cloud environment becomes a natural choice for short-term readiness. Later, for uniformity of organisational policies and a unified platform, an organisation can move to a single cloud provider (strategic cloud platform).
3. It offers availability across multiple cloud service providers, which reduces risks associated with security, scalability and availability.
4. Multi-cloud management is a great option for disaster recovery (DR) sites, where one cloud provider acts as the primary site and another acts as a secondary one to ensure high-availability of applications at any given time.

When large enterprises that span across multiple geographies want to migrate to the cloud, choosing a single cloud service provider is a big challenge due to the geographical diversity and related reasons. In such a scenario, choosing a multi-cloud strategy is the natural choice for enterprise cloud adoption. As per a Gartner survey, about 81 per cent of large enterprises are using more than one cloud service provider and there are multiple reasons for that, such as:

1. Avoiding vendor lock-in
2. Cost optimisation for apps depending on the geographical diversity
3. Usage of native services for different applications provided by different cloud service providers

According to a 2020 report from Gartner, cloud management comprises seven functional areas:

- Infrastructure and service provisioning, and orchestration
- Service enablement and faster adoption to cloud services
- Inventory and classification of cloud services based on complexity and priority
- Identity, security and compliance to handle application/ data/platform security — a compliance framework